1.#include<string.h>

(1M)

```
#include<stdio.h>
int main()
{
int a[]={11,22,33,44,55,66};
char s[]="vector";
int n1,n2;
n1=sizeof(a)/sizeof(a[0]);
n2=sizeof(s)/sizeof(s[0]);
printf("%d %d %lu",n1,n2,strlen(s));
}
```

A. 6 7 6 B. 6 6 6 C. None D. 6 7 7

2.#include<stdio.h>

(1M)

```
int main()
{
int len=4;
char *st="12345678";
st = st + len;
printf(" %c" ,*st ) ;
}
```

A. 6 B. 5 C. 4 D. Run time error

3.#include<stdio.h>

(1M)

```
int main()
{
```

```
const int x=5;
const int *ptrx;
ptrx = &x;
*ptrx = 10;
printf("%d", x);
return 0;
}
```

A. 5 **B. error** **C. 10** **D. garbage**

(1M)

4.

```
#include<stdio.h>
#include<string.h>
int main()
{
char s1[15]="visualstudio";
char *p="s";
char *q=strstr(s1,p);
printf("%s",q);
}
```

A. sualstudio **B. studio** **C. visualstudio** **D. error**

(1M)

5.

```
#include<stdio.h>
int i=15;
int main()
{
int i=20;
printf("%d",i);
}
```

A.15 **B.20** **C. Error** **D. Garbage value**

(1M)

6.#include<stdio.h>

#include<stdlib.h>

#include<string.h>

int main()

{

char *p=NULL;

p=malloc(8);

strcpy(p,"vector");

p=realloc(p,15);

strcat(p," india");

puts(p);

free(p);

return 0;

}

A. vector india

B. india

C. segmentation fault

D. vector

(1M)

7.What is the output for this program....if input is ./a.out sun run

#include<stdio.h>

int main(int argc,char *argv[ ])

{

```
printf("%s %s",argv[0],argv[1]);
}
```

A. ./a.out sun B. Depends on the compiler

C. Segmentation fault D. sun run

8. Predict the output of the following code

```
#include<stdio.h>
int main()
{
printf("%d\n",++EOF);
}
```

A. EOG B. 0 C. -1 D. None

9. Predict output of the following

```
#include<stdio.h>
union tag
{
unsigned int i;
char c[4];
}var={0x61626364};
int main( )
{
printf("%c",var.c[1]);
```

Module : C / MAX Marks : 25

}

A. a B. b C. c D. None

10. Write C Program to Print the Binary Format of a integer (3M)

11. Complete the following macro definitions to manipulate bits in a given integer. (3M)

```
#include <stdio.h>
// Macros to manipulate bits...fill in the macros

#define SET_BIT(x, n)      x|<<n

#define CLEAR_BIT(x, n)    x&~(1<<n)

#define CHECK_BIT(x, n)    x>>n&1

int main() {
   unsigned int number = 0;
   int bit_position = 3;
   SET_BIT(number, bit_position);
printf("After setting bit %d, number = %u\n", bit_position, number);
   CLEAR_BIT(number, bit_position);
printf("After clearing bit %d, number = %u\n", bit_position, number);
   int bit_status = CHECK_BIT(number, bit_position);
printf("The status of bit %d is: %d\n", bit_position, bit_status);
   return 0;
}
```

***

12.Implement the function =>remove_extraspace_between_words (5M)

This function should remove all extra spaces between words, ensuring only single spaces separa words.

Example:

Input: "This is a test"

Output: "This is a test"

```
#include <stdio.h>
#include <string.h>
void remove_extraspace_between_words(char *p)
{
write code here....
.....
}
int main() {
char str[50];
printf("Enter string: ");
use fgets() to take input
remove_extra_spaces_middle(str);
```

Vector Shortlisting Test – August 2024

Module : C / MAX Marks : 25

```
printf("Updated string: %s\n", str);
return 0;
}
```

.....***.....

13. Write a C program that hides a specific substring within a main string by replacing the characters of the substring with asterisks (*). (5M)

Implement the hide_substring function as described above.

Example:

Input:

- Main string: "Hello, this is a simple test."
- Substring: "simple"

Output:

- "Hello, this is a ****** test."

```
#include <stdio.h>
#include <string.h>
void hide_substring(char *main_str, const char *sub_str)
{
write code here....
.....
```

```
}
int main() {
    char main_str[100], sub_str[50];
  printf("Enter the main string: ");
    //use fgets() to take input
  printf("Enter the substring to hide: ");
    //use fgets() to take input
    // Call the function to hide the substring
hide_substring(main_str, sub_str);
printf("Modified string: %s\n", main_str);
  return 0;
}
```

.....ALL THE BEST.....

1.scope resolution operator is used

a.In case of multiple Inheritance

b.To define a function outside a class / namespace

c.To access a global variable when there is a local variable with same name

d.all of these

2.Which of the following is false about the references in c++?

a.Once reference is refered to variable , it cannot be rerefered to another variable

b.Reference must intialized when declared

c.We cannot create a reference to a constant value

d.Reference cannot be NULL

3.

```
#include <iostream>
#include <string>
using namespace std;
namespace A{

int var = 10;
}
```

```
namespace B{
int var = 5;
}
int main()
{
int var = 20;
using namespace B;
cout<<var;
}
```

a.error

b.5

c.20

d.10

4.What is the correct syntax of defining a namespace?

a. namespace name{}

b. Namespace name{};

c. typedef namespace name{} NAME

d. namespace name{};

5. How to print the value of the i variable inside namespace B?

```
namespace A{
int var = 10;
namespace B{
int i = 15;
```

}

}

a.cout<<A::i;

b.cout<<i;

c.cout<<A::B::i;

d.cout<<B::i;

6. The C++ language is _____ object-oriented language.

a. None of the above

b. Not Object oriented

c. Semi Object-oriented or Partial Object-oriented

d. Pure Object oriented

7.How ismemory allocated for an array dynamically in C++?

a. int arr = new int[10];

b. int* arr = allocint[10];

c. int* arr = new int[10];

d. int arr[10] = new int;

8.What does the 'delete[]' operator do in C++?

a. Deletes an array from the stack

b. Deletes a pointer

c. Frees memory allocated for an array

d. Frees memory allocated for a single variable

9.what is the size of empty structure in c++?

a.0 byte

b.4 byte

c. none of these

d.1 byte

10.#include<iostream>

```
using namespace std;
main() {
 struct ABC { int abc:2; }KK;
KK.abc=3;
cout<<KK.abc<<endl;
}
```

a.2

b.-1

c.-3

d.3

11.The function that can not modify the object that invokes it, is said to be?

a.virtual function

b.const member function

c. friend function

d. inline function

12.Size of an object is equal to_____

a.none of these.

b.size of data members only

c.sum of sizes of data members and member functions

d.size of members functions only

13.What is a copy constructor?

a. A constructor to check the whether to objects are equal or not

b. A constructor to kill other copies of a given object

c. A constructor to initialize an object with the values of another object

d. A constructor that allows a user to move data from one object to another

14.Why constructors are efficient instead of a function init() defined by the user to initialize the data members of an object?

a.Because user may forget to call init() using that object leading segmentation fault

b. Because user may forget to define init() function

c. Because user may call init() more than once which leads to overwriting values

d. All of the mentioned

15.we can use const on

a. objects

b. fuction arguments

c. normal variables

d. all of these

16.this pointer can be used inside in static member functions(True/False)? False

17.Which of the following operators should be preferred to overload as a global function rather than a member method?

a. Insertion Operator <<

b. Prefix++

c. Comparison Operator

d. Postfix ++

18.When overloading unary operators using Friend function, it requires_____ argument/s.

a. Zero

b. One

c. Two

d. None of these.

19.When the inheritance is private, the private methods in base class are _____ in the derived class

a. Inaccessible

b. Accessible

c. Protected

d. Public

Module : CPP / M.Sc. Marks : 25

20.Which concept is not available in C++?

a.Virtual constructor

b.Virtual destructor

c.Virtual function

d.Virtual Table

21.#include<iostream>

using namespace std;

```
class A
{
virtual void test1() {
cout<< "test1 in A" <<endl;
}

};
class B:public A
{
public:
void test1() {
cout<< "test1 in B" <<endl;
}
};
```

Module : CPP / MAX Marks : 25

```
main()
{
A *aptr;
B b1;
aptr=&b1;
aptr->test1();
}
```

a.error

b.test1 in B

c.test1 in A

test1 in B

d.test1 in A

22.It is necessary that the virtual function overridden in the derived class must have the same signature?(TRUE/FALSE)?

True

23.exception handling is used to handle

a. compile time errors

b. run time errors

c. none of these

d. both runtime and compile time errors

Module : CPP / MAX Marks : 25

24.What is the validity of template parameters?

a. inside that block only

b. inside the class

c. whole program

d. inside the main class

25.Which header file is required to use file I/O operations?

a.<ifstream>

b.<ostream>

c.<fstream>

d.<iostream>

1. The ARM7TDMI-S is the type of _____ & _____.

A) Harvard & CISC B) Harvard & RISC C) Von-Neumann & CISC D) Von-Neumann &RISC

2. Size of the Address Bus for ARM 7 Core is _____, so it can address _____ of memory, with and address ranges _____ to _____.

A) 16-bit, 2GB, 0x00000000, 0xFFFFFFFF B) 64-bit, 4GB, 0x00000000, 0xFFFFFFFF

C) 32-bit, 4GB, 0x00000000, 0xFFFFFFFF D) 32-bit, 2GB, 0x00000000, 0xFFFFFFFF

3. #define WRITEBIT(WORD,BITPOS,BIT) (_____?SETBIT(WORD,_____):CLRBIT(WORD,_____))

A) BIT, BITPOS, BIT B) BIT, BITPOS, BITPOS

C) BIT, BIT, BIT D) BITPOS, BITPOS, BITPOS

4. What is the size of Flash ROM & RAM in the LPC2129 Microcontroller?

A) 512KB & 32KB B) 32KB & 512KB C) 256KB & 16KB D) 16KB & 256KB

5. IOPIN register is used for _____ operations.

A) input B) output C) Both A&B D) None of the mentioned

6. In LPC2129, there are _____ SFRs in the Pin Connect Block & _____ SFRs in the GPIO Peripheral Block.

A). 3, 4 B) 3, 8 C) 8, 3 D) 4, 3

7. For Arranging 20 Push Button Switches in the form of Keypad Matrix, the possible combination of rows & columns are _____, with the least possible pin count.

A) 2X10 B) 4X5 C) 10X2 D) None of the mentioned

8. The _____ is used for Configuring 1 out of 4 functionalities for each of the 46 I/O pin in the 64pin QFF.

A) GPIO Peripheral Block B)VLSI Peripheral Blocks

C) Phase Locked Loop & D) Pin Connect Block

9. The number of I/O pins maximum required to directly operate a 2 - multiplexed 7 segment display unit with support for decimal point leds also are

A) 8 B) 9 C) 10 D) 11

10. In UART protocol data transfer or receive always_____first.

A) LSB B) MSB C) LSB or MSB D) None of the mentioned

11. UART's Protocol is an _____ serial communication protocol whereas _____ & _____ are synchronous serial communication protocols.

A) synchronous full duplex, i2c, spi B) synchronous half duplex, i2c, spi

C) asynchronous half duplex, i2c, spi D) asynchronousfull duplex, i2c, spi

Module: ARM / MAX Marks: 25

112. What is interrupt latency?

A) It is the time that elapses from when an interrupt is generated to when the source of the interrupt is serviced

B) It is defined as the time it takes from the assertion of the interrupt request signal by the interrupting device until the start of the execution of the first useful instruction in the ISR

C) It is the time taken by the CPU to return from ISR to the next instruction of the interrupted function

D) All of the mentioned

113. Among all I2C bus events, the bus event(s) _____ is/are one(s) in which SDA line can transits while SCL line remains high.

A) Start B) Repeated Start C) Stop D) All of the mentioned

114. I2C Protocol is Multi Master Slaves Communication Protocol with a Slave be accessed using _____.

A) 10-bit Slave Address B) Either 7-bit/10-bit Slave Address

C) 7-bit Slave Address D) Both 7-bit & 10-bit Slave Address

115. The LCD Units using HD44780 controller, irrespective of Display Screen/Panel Sizes has _____ pin outs.

A) 14 B) 16 C) 8 D) 3

116. In Rs-232 voltage range is _____ to ___ V.

A) +3v to +25v B) -3v to -25v C) -25v to +25v D) -3v to +3v

117. For the function definition given below, with comments, is used for _____.

Module: ARM / MAX Marks: 25

```
void i2c_event_function(void)
{
        //SDA_HIGH
        //SCL_HIGH
        //SDA_DIR_OUT
        //SCL_DIR_OUT
        //SDA_LOW;
}
```

A) start event B) repeated start event C) stop event D) None of the mentioned

18. What is the size of the SPI Master shift register in LPC2129 controller?

A) 1 Byte B) 16 Bits C) 2 Bytes D) 4 Bytes

19. The SPI protocol is _____ & _____ communication protocol.

A) Multimaster & half duplex B)Multimaster & full duplex

C) Multislave & half duplex D) Multislave & full duplex

20. The SPI Protocol support 4 modes of communication depending on "clock" signal expressed as "Clock Polarity (CPOL)" & "Clock Phase (CPHA)".

A) TRUE B) FALSE

21. In the data-frame of CAN, the value of RTR bit is _____.

A) Dominant B) Recessive C) Both A&B D) None of the mentioned

22. I2C Protocol Uses Wired - And - Logic for Bus Arbitration, Clock Synchronization & Clock Stretching.

A) TRUE B) FALSE

Module: ARM / MAX Marks: 25

```
void i2c_event_function(void)
{
        //SDA_HIGH
        //SCL_HIGH
        //SDA_DIR_OUT
        //SCL_DIR_OUT
        //SDA_LOW;
}
```

A) start event B) repeated start event C) stop event D) None of the mentioned

18. What is the size of the SPI Master shift register in LPC2129 controller?

A) 1 Byte B) 16 Bits C) 2 Bytes D) 4 Bytes

19. The SPI protocol is _____ & _____ communication protocol.

A) Multimaster & half duplex B)Multimaster & full duplex

C) Multislave & half duplex D) Multislave & full duplex

20. The SPI Protocol support 4 modes of communication depending on "clock" signal expressed as "Clock Polarity (CPOL)" & "Clock Phase (CPHA)".

A) TRUE B) FALSE

21. In the data-frame of CAN, the value of RTR bit is ____.

A) Dominant B) Recessive C) Both A&B D) None of the mentioned

22. I2C Protocol Uses Wired - And - Logic for Bus Arbitration, Clock Synchronization & Clock Stretching.

A) TRUE B) FALSE

Module: ARM / MAX Marks: 25

23. Reflections in CAN networks are avoided by termination of the bus ends.

A) TRUE B) FALSE

24. Which component of a CAN node handles the CAN protocol?

A) CAN Transceiver B) CAN Controller C) MCU D) None of the mentioned

25. For the function definition given below the identifier "i2c_function" can be renamed as _____.

```
void i2c_function(void)
{
        //Cfg p0.2 pin for SCL function using PINSEL0
        //Cfg p0.3 pin for SDA functions using PINSEL0
        //Cfg Speed for I2C Serial Communication using I2SCLL, I2SCLH
        //Enable I2C Peripheral for Communication using I2CONSET
}
```

A) i2c_start B) i2c_init C) i2c_stop D)i2c_ack

1. int main()

```
{   int ret=fork();
    if(ret>0)
    {
    printf("i am in parent process pid :%d  ppid:%d\n",getpid(),getppid());
    exit(0);
    }
    else
    {
    printf("i am in child process pid :%d  ppid:%d\n",getpid(),getppid());
        sleep(10);
    }
}
```

In the above code, which process it will be zombie process or orphan process? Orphan process

2.Virtual memory is _____.

a. An illusion of extremely large main memory

b. An extremely large secondary memory

c. An extremely large main memory

d. A type of memory used in super computers.

3.Thread synchronization is required because

a. all threads of a process share the same address space

b. all threads of a process share the same global variables

c. all threads of a process can share the same files

d. all of the mentioned

Module : LINUX / MAX Marks : 25

4. In Linux, what is the primary difference between a thread and a process?

a. Threads share the same memory space, while processes have separate memory spaces.

b. Threads run in the kernel mode, while processes run in user mode.

c. Threads have higher priority than processes.

d. Threads can be created without the need for system calls.

5. A circular queue is the most appropriate data structure for _____ scheduling.

a. Round Robin

b. First Come First Serve

c. Shortest Job First

d. None of the above

6. The rmcommand in Linux will permanently delete files and directories without moving them to any kind of trash or recycle bin.

a. True

b. False

7.In Linux, what is the purpose of the umask command?

a. To set the default file permissions for newly created files.

b. To mask certain characters in filenames.

c. To move files between directories.

d. To unmount file systems.

8. Consider relevant header files are included.

```
int main()
{
    int ret1,ret2,ret3;
    ret1=open("textfile1.txt",O_RDONLY); //Consider textfile1.txt is Not present in current folder.
    ret2=open("textfile2.txt",O_RDONLY);// /Consider textfile2.txt is present in current folder.
    ret3=open("textfile3.txt",O_RDWR|O_CREAT,0660);//Consider textfile3.txt is Not present in current
                                                    folder.

  printf("%d %d %d\n",ret1,ret2,ret3);   // -1 1 1
  return 0;
}
```

What will be output for the given code?

Module : LINUX / MAX Marks : 25

9.In Linux, what is the significance of a file with the permission set to -rwxr-xr--?

a. The owner can read and write, the group can read, and others have no permissions.

b. The owner can read, write, and execute; the group can read and execute; others can read only.

c. The owner can read and write, the group can execute, and others can read only.

d. The file is executable by the owner, writable by the group, and readable by others.

10.

Process1.c

```
#include<stdio.h>
int main()
{
char *p[]={"vector","india","chennai"};
printf("%s %s %s \n ",*p+1,*p+2,*p+3);
puts("process1");
return 0;
}
Gcc Process1.c  -o  p.exe
```

Process2.c

```
#include<stdio.h>
int main()
{
    char
*args[]={"VST","TEST",NULL};
    puts("process 2");
    execv("./p.exe",args);
    puts("process 2");
}
Gcc Process2.c
```

What will be output for the process2.c? process 2
VST TEST

11.Which of the following is true about hard links in Linux?

a).They can be created across different filesystems.

b. Deleting the original file also deletes the hard link.

c. They point to the same inode as the original file.

d. They store the path to the original file.

12. Which of the following IPC mechanisms can be used for communication between unrelated processes in Linux?

a. Pipes

b. Named Pipes (FIFOs)

c. Signals

d. Memory-mapped files

13.In Linux, what is the role of a semaphore in IPC?

a. To send messages between processes.

b. To synchronize access to shared resources.
c. To map files into memory.
d. To establish a communication channel between unrelated processes.

14.//Consider relevant header files are included.

```
int var=100;
int main()
{   int ret;
    ret=fork();//lets consider fork call is success.
    switch(ret)
    {
        case -1:var++;
        case 0: var++;
        default:var++;
    }
    if(ret==0)
        printf("in child...var=%d\n",var);   // 100
    else
        printf("in parent...var=%d\n",var);  // 101
}
```

What will be value of var in parent and in child?

15.Which command can be used to view the IPC facilities used by the system, such as message queues, shared memory, and semaphores?

a. ipcs
b. ipcrm
c. lsof
d. ps

16. what is the signalnumber for SIGSEGV signal

a. 3

Module : LINUX / MAX Marks : 25

b. 10

c. 12

d. 11

17.//Consider relevant header files are included.

```
int main()
{
    printf("Hello\n");
    execlp("date","date",NULL);
    fork();
    fork();
    printf("Bye\n");
    return 0;
}
```

how many times Bye will be printed? A:- 0 times

18.What is the purpose of the SIGCHLDsignal in Linux?

a. It notifies a parent process that a child process has stopped or terminated.

b. It kills the child process when received by the parent process.

c. It is used to stop a process temporarily.

d. It continues a stopped process.

19.Which of the following signals cannot be caught, blocked, or ignored?

a. SIGTERM

b. SIGKILL

c. SIGINT

d. SIGQUIT

20._____ does the job of allocating a process to the processor.

a. Short term scheduler

b. Long term scheduler

c. Medium term scheduler

d. Dispatcher

21.Which of the following is an advantage of using shared memory for inter-process communication in Linux?

a. It is slower than message passing.

b. It allows processes to share large amounts of data efficiently.

c. It is limited to communication between parent and child processes.

d. It does not require synchronization mechanisms.

22.The mechanism that bring a page into memory only when it is needed is called _____

a. Segmentation

b. Fragmentation

c. Demand Paging

d. Page Replacement

23.What does the sudo command do in Linux?

a. It changes the file permissions.

b. It allows a permitted user to execute a command as the superuser.

c. It deletes files permanently.

d. It suspends a process.

24.What is the primary disadvantage of using shared memory for IPC?

a. It is slower than other IPC mechanisms.

b. It requires explicit synchronization mechanisms like semaphores.

c. It can only be used for communication between related processes.

d. It has a limited message size.

25.Which of the following commands is used to terminate a process by its process ID (PID) in Linux?

a. kill

b. terminate

c. end

d. stop

1. Identify the class of following IP address: 4.5.6.7.

A) Class A B) Class B C) Class C D) Class D

2. The loop-back address is used to send a packet from the ____ to ____ on the network.

A) Host; all other hosts B) Router; all other hosts C) Host; specific host D) Host; itself

3. What destination address can be used to send a packet from a host with IP address 188.1.1.1 to all hosts on the network?

A) 188.0.0.0 B) 0.0.0.0 C) 255.255.255.255 D) 188.1.255.255

4. Repeaters function in ___ layer of OSI?

A) Physical B) Data link C) Network D) a and b

5. As the data packet moves from lower to upper layers, headers are___

A) Added B) Subtracted C) Rearranged D) Modified

6. What is the main function of the transport layer?

A). Node-to-node delivery B). Process to process message delivery

C). Synchronization D). Updating and maintenance of routing tables

7. IPv6 has ___ bit addresses.

A) 32 B) 64 C) 128 D) Variable

8. When a host on network A sends a message to a host on network B, which address does the router look at?

A) Port B) IP C) Physical D) None of the above

9. Which of the following is not an IPC event?

A) pipe B) Message queue C) fork D) Socket

10. Socket is using

A) Port number B) Interrupt number C) Memory address D) All

11. What kind of byte-ordering is following in NETWORK

a) little endian b) big endian c) depends on n/w d) none of these

12. which of the following is true about bind()

a) bind is must in client process b) use to accept incoming connection

c) Third argument in bind is a pointer d) use to assign address to created socket

13. A broadcast message is an example of which dialog control method

a) Half Duplex b) Base band c) Broad band d) Simplex

14. Which of the following is Class C IP address?

a) 10.10.14.118 b) 135.23.112.57 c) 191.200.199.199 d) 204.67.118.54

15. What is the first octet range for a class A IP address?

a) 1 – 126 b) 1 – 128 c) 1 – 198 d) none

16. How many bits are there in an IP address V4 ?

a) 16 b) 32 c) 64 d) 128

17. Maximum port number allowed in socket is

a) upto 4 GB b) upto 65535 c) no limit d) 255

18. Accept system call returns

a) a new socket id on success b) replaces the existing socket id on failure

c) a and b d) none

19. What are the port numbers of the following SMTP, FTP, TFTP?

a) 21,25,69 b) 25,21,69 c) 69,21,25 d) none

20. What is the responsibility of TCP ?

a) Multiplexing b) reliability c) Flow control d) all

21. Which function is used to SEND data in connection-less Host

a) sendto b) send c) write d) any of the above

Vector Short Listing Test – AUGUST 2024

Module : TCPIP / MAX Marks : 25

22. The so-called "well-known" TCP and UDP port numbers fall into which numeric range?

a) 0 to 999 b) 0 to 1000 c) 0 to 1023 d) 0 to 1024

23. Which of the following IPv4 addresses is the *loopback address*?

a) 0.0.0.0 b) 10.0.0.1 c) 127.0.0.1 d) 255.255.255.255

24. The maximum size of an IP datagram (including the header) is _____

a) 1,024 bytes b) 1,500 bytes c) 4,096 bytes d) 65,535 bytes

25. what is the use of port number???

a) to identify the app b) to identify the PC c) to identify the host d) to identify the network